CH 27 — MAGNETIC FORCE

$\vec F=q\vec v\times\vec B$; $|F|=qvB\sin\theta$
$\vec F$ wire: $\vec F=I\vec L\times\vec B$; $|F|=BIL\sin\theta$
$\Phi_B=\int\vec B\cdot d\vec A=BA\cos\theta$ [$T\cdot m^2$]

Circular motion

$r=mv/(qB)$; $T=2\pi m/(qB)$; $\omega=qB/m$
$v=qBr/m$; $KE=q^2B^2r^2/(2m)$

★ Velocity selector / Mass spec

★ **Vel sel:** $v=E/B$ ($qE=qvB \leftrightarrow$ undeflected)
★ $V=Ed$ across plates
★ **Mass spec:** $v=\sqrt{2qV/m}$; **$m/q=B^2R^2/(2V)$**
★ **Free-fall into B:** $v=\sqrt{2gh}$; $F=qvB$

★ Magnetic Torque on Loop

$\vec\mu=NIA\cdot\hat n$ (mag dipole moment); $\hat n$ from RHR around I
$\vec\tau=\vec\mu\times\vec B$; $|\tau|=\mu B\sin\theta=NIAB\sin\theta$
$U=-\vec\mu\cdot\vec B=-\mu B\cos\theta$
★ Stable equilib: $\vec\mu\parallel\vec B$ ($\theta=0$, U min). Unstable: antiparallel ($\theta=\pi$).

CH 28 — AMPERE'S LAW

$\oint B\cdot dl=\mu_0 I_{encl}$
Biot–Savart: $dB=(\mu_0/4\pi)\cdot(I\,d\vec l\times\hat r)/r^2$
Long wire: $B=\mu_0I/(2\pi r)$
Inside solid wire (r<R): $B=\mu_0Ir/(2\pi R^2)$
Solenoid: $B=\mu_0nI$ ($n=N/L$)
Toroid in: $B=\mu_0NI/(2\pi r)$; **Out:** $B=0$
Ctr of circ loop: $B=\mu_0I/(2R)$
Moving pt charge: $B=(\mu_0/4\pi)\cdot qv\sin\theta/r^2$
▶ **Multi-I sign:** CCW path → + out; CW → + in

Parallel wires

$F/L=\mu_0I_1I_2/(2\pi d)$. Same dir→ATTRACT; opp→REPEL

★ Cheats

★ **Coax** (I_1 in, I_2 out same dir): $a<r<b$: $B=\mu_0I_1/(2\pi r)$; $r>c$: $B=\mu_0(I_1+I_2)/(2\pi r)$. Opp&equal: B=0 outside.
★ **3 ∥ wires (eq d):** outer $F/L=\mu_0I^2/(4\pi d)$; middle (sym)=0
★ **Square loop side a near wire** (ctr d, ∥): $F=\mu_0I_1I_2a^2/[2\pi(d^2-a^2/4)]$

CH 29 — FARADAY & INDUCTION

$\varepsilon=-d\Phi_B/dt$; $\oint E\cdot dl=-d\Phi_B/dt$
$\Phi_B=BA\cos\theta$ ($T\cdot m^2$, not Wb!)
Lenz: induced I opposes $\Delta\Phi$

★ EMF when B(t) and/or θ(t) vary

▶ **B(t) varies, θ const:** $\varepsilon=-A\cos\theta\cdot(dB/dt)$
▶ **θ(t) varies, B const:** $\varepsilon=BA\sin\theta\cdot(d\theta/dt)$
▶ **Both vary:** $\varepsilon=-A[\cos\theta\cdot dB/dt-B\sin\theta\cdot d\theta/dt]$
▶ **Rotating coil:** $\varepsilon=NBA\omega\sin(\omega t)$; $\varepsilon_{max}=NBA\omega$

Motional EMF — Slide-wire

$\varepsilon=BLv$; $I=BLv/R$
F to push: $F=B^2L^2v/R$; $P=(BLv)^2/R$
★ Loop entering/exiting B: $I=BLv/R$ during entry & exit (opp dirs); I=0 fully inside

★ Induced E from changing B

★ **Solenoid w/ dI/dt:** $r<R_{sol}$: $E=(\mu_0nr/2)(dI/dt)$; $r>R_{sol}$: $E=(\mu_0nR^2/(2r))(dI/dt)$; r=0: E=0
★ **Cyl w/ $B(t)=B_0+B_1t$** outside: $|E|=B_1A/(2\pi r)$

Displacement current

$I_D=\varepsilon_0(d\Phi_E/dt)$; $j_D=\varepsilon_0(dE/dt)$
Ampere–Maxwell: $\oint B\cdot dl=\mu_0(I_C+I_D)$
★ Charging cap: r<R: $B=\mu_0I_C\cdot r/(2\pi R^2)$; r>R: $B=\mu_0I_C/(2\pi r)$

CH 30 — INDUCTANCE & RLC

$\varepsilon_L=-L(dI/dt)$; $L=N\Phi_B/I$ [H]
Solenoid: $L=\mu_0N^2A/\ell$; **Toroid:** $L=\mu_0N^2A/(2\pi r)$
$U=\frac{1}{2}LI^2$; $u_B=B^2/(2\mu_0)$
Mutual ind: $M=N_2\Phi_{21}/I_1$; $\varepsilon_2=-M(dI_1/dt)$

RL Circuit | τ=L/R

▶ **Energizing:** $i(t)=(\varepsilon/R)[1-e^{-Rt/L}]$
▶ **Decay:** $i(t)=I_0e^{-Rt/L}$
▶ t=0: i=0 (inductor blocks); t=∞: $i=\varepsilon/R$
★ Find L from decay: $L=Rt/\ln(I_0/I)$

LC Oscillation

$\omega_0=1/\sqrt{LC}$; $f=1/(2\pi\sqrt{LC})$
$q(t)=Q\cos(\omega t)$; $i(t)=-Q\omega\sin(\omega t)$
Energy: $U=Q^2/(2C)=\frac{1}{2}LI^2_{max}$

★ Damped LRC (series)

★ $\omega'=\sqrt{1/LC-R^2/(4L^2)}$
★ Amp decay: $A(t)=A_0e^{-Rt/(2L)}$; t to fraction f: $t=-(2L/R)\ln(f)$
★ Critical: $R_c=2\sqrt{L/C}$. $R<R_c$ under; $R>R_c$ over

CH 31 — AC CIRCUITS

$V(t)=V_{max}\cos\omega t$; $I(t)=I_{max}\cos(\omega t-\varphi)$
$V_{rms}=V_{max}/\sqrt{2}$; $I_{rms}=I_{max}/\sqrt{2}$
R: in phase, $V_R=IR$
L: V leads 90°, $X_L=\omega L$, $V_L=IX_L$
C: V lags 90°, $X_C=1/(\omega C)$, $V_C=IX_C$
"ELI the ICE man" — E leads I in L; I leads E in C

Series LRC

$Z=\sqrt{R^2+(X_L-X_C)^2}$; $I=V/Z$
$\tan\varphi=(X_L-X_C)/R$
Resonance: $X_L=X_C$; $\omega_0=1/\sqrt{LC}$; Z=R
$P_{avg}=V_{rms}\cdot I_{rms}\cdot\cos\varphi=I^2_{rms}\cdot R$

★ Phasor Diagram Tips

★ V_R along I (in phase); V_L straight up (+90°); V_C straight down (−90°). V_{source} = vector sum.
★ $|V_{source}|^2=V_R^2+(V_L-V_C)^2$

★ Transformer (ideal)

★ **$V_2/V_1=N_2/N_1$** (turns ratio)
★ **$I_2/I_1=N_1/N_2$** (inverse, conserves P)
★ $P_1=P_2\to V_1I_1=V_2I_2$ (ideal)
★ Step-up: $N_2>N_1\to V\uparrow, I\downarrow$. Step-down: opposite.

CH 32 — EM WAVES

▶ **$E=E_m\sin(kx-\omega t)$; $B=B_m\sin(kx-\omega t)$**
▶ **$E_m/B_m=c$; $E(t)/B(t)=c$**
▶ **$c=1/\sqrt{\mu_0\varepsilon_0}=f\lambda=\omega/k$**
▶ **$k=2\pi/\lambda$; $\omega=2\pi/T=2\pi f$**
▶ **$\vec S=(1/\mu_0)(\vec E\times\vec B)$**; $|S|=EB/\mu_0=E^2/(c\mu_0)=cB^2/\mu_0$
▶ **$I=S_{avg}=E_mB_m/(2\mu_0)=\frac{1}{2}c\varepsilon_0E_m^2=E_m^2/(2\mu_0c)$**
Rad pressure: Absorb: P=I/c. Reflect: P=2I/c
$\vec E\times\vec B$ → propagation (RHR). Spectrum (low→high f): Radio < μW < IR < Vis(400-700nm) < UV < X < γ
In medium: v=c/n (n=index of refraction)

★ MAXWELL'S 4 EQUATIONS

1. **Gauss (E):** $\oint E\cdot dA=Q_{encl}/\varepsilon_0$
2. **Gauss (B):** $\oint B\cdot dA=0$ (no monopoles)
3. **Faraday:** $\oint E\cdot dl=-d\Phi_B/dt$
4. **Ampere-Maxwell:** $\oint B\cdot dl=\mu_0(I_C+\varepsilon_0\,d\Phi_E/dt)$

★ RIGHT-HAND RULE GUIDE

F=qv×B (charge): fingers along v, curl to B, thumb=F (flip if q<0)

B from wire: thumb along I, fingers curl in dir of B

Ampere loop: fingers along path, thumb = + I direction

Loop μ: fingers curl with I, thumb = μ direction

EM wave: fingers E, curl to B, thumb = propagation

Lenz: Find dΦ direction, induced B opposes it, then RHR for I

★ COMMON PITFALLS

⚠ Φ_B in $T\cdot m^2$ (NOT Wb) per prof
⚠ RC charging is $1-e^{-t/RC}$; discharging is $e^{-t/RC}$. Don't swap.
⚠ RL: at t=0 inductor BLOCKS (i=0). Capacitor at t=0 is a wire (V=0).
⚠ "Toroid" r is from CENTER (not from wire).
⚠ Cap disconnected → Q const, E unchanged when d changes. Connected → V const.
⚠ cos vs sin in Φ: cosθ when θ from normal; sinθ if from plane.
⚠ In LRC: $V_C+V_L+V_R\neq V_{source}$ (algebraic). Use phasor sum.
⚠ In Ampere: r is from current axis, not anywhere else.
⚠ Mass spec: V is accelerating voltage, not vel-selector V.
⚠ $\tau_{RC}=RC$, $\tau_{RL}=L/R$ — DIFFERENT formulas, easy to swap.
⚠ "Loop fully inside B" → no dΦ → I = 0 (only changing flux induces).
⚠ Direction of induced I: use Lenz, NOT the sign in Faraday's eq.

🎯 EXAM TRIAGE — KEYWORD → FORMULA

"insulating sphere E in/out" → kQr/R^3 vs kQ/r^2
"infinite line/wire E" → $\lambda/(2\pi\varepsilon_0r)$
"infinite sheet E" → $\sigma/(2\varepsilon_0)$
"conductor surface E" → σ/ε_0
"point charge E" → kQ/r^2
"point charge V" → kQ/r
"ring on axis E" → $kqz/(z^2+a^2)^{3/2}$
"semicircle ±Q" → $4kQ/(\pi a^2)$
"square alt ±q" → $4\sqrt{2}\,kq/a^2$
"cavity in sphere" → $\rho h/(3\varepsilon_0)$
"flux thru hemisphere" → $q/(2\varepsilon_0)$
"Φ pt charge full sphere" → q/ε_0
"V from line of charge" → $2k\lambda\ln(r_0/r)$
"work to move charge" → $W=q\Delta V$
"speed gained from V" → $v=\sqrt{2qV/m}$
"E between plates" → V/d
"force on charge in E" → F=qE
"dipole torque" → $pE\sin\theta$
"dipole U" → $-pE\cos\theta$
"cap C from area" → ε_0A/d
"cap energy" → $\frac{1}{2}CV^2=Q^2/(2C)$
"cap energy density" → $\frac{1}{2}\varepsilon_0E^2$
"cap with dielectric" → $C=\kappa\varepsilon_0A/d$
"cap disconn., move plates" → Q const, U∝d
"cap stays connected" → V const, U∝1/d
"caps in series" → $1/C_{eq}=\Sigma 1/C_i$
"caps in parallel" → $C_{eq}=\Sigma C_i$
"two C in series" → $C_1C_2/(C_1+C_2)$

"equiv resistance" → reduce groups
"two R parallel" → $R_1R_2/(R_1+R_2)$
"n equal R parallel" → R/n
"power in resistor" → $I^2R=V^2/R$
"power from battery" → εI
"battery terminal V" → ε−Ir
"two batteries opposing" → $I=|\varepsilon_1-\varepsilon_2|/\Sigma R$
"two batteries aiding" → $I=(\varepsilon_1+\varepsilon_2)/\Sigma R$
"wire R from L,A" → R=ρL/A
"drift velocity" → $I=nqv_dA$
"current density J" → I/A
"τ for RC" → RC
"τ for RL" → L/R
"RC voltage at time t" → $V_C=\varepsilon(1-e^{-t/RC})$
"RC current" → $(\varepsilon/R)e^{-t/RC}$
"RC initial (t=0)" → $V_C=0$, $V_R=\varepsilon$, $i=\varepsilon/R$
"RC final (t=∞)" → $V_C=\varepsilon$, $V_R=0$, i=0
"RC discharge V" → $V_0e^{-t/RC}$
"time to charge to half" → t=RC ln 2
"time to decay to 1/e" → t=τ
"loop entering B" → I=BLv/R
"slide wire EMF" → ε=BLv
"force to push rod" → B^2L^2v/R
"power dissipated rod" → $(BLv)^2/R$
"undeflected ion" → v=E/B
"mass spec m/q" → $B^2R^2/(2V)$
"speed after accel V" → $\sqrt{2qV/m}$
"radius circ motion B" → r=mv/(qB)

"period in B-field" → $2\pi m/(qB)$
"cyclotron freq" → $qB/(2\pi m)$
"force ∥ wires/length" → $\mu_0I_1I_2/(2\pi d)$
"toroid B inside" → $\mu_0NI/(2\pi r)$
"toroid B outside" → 0
"solenoid B" → μ_0nI
"moving charge B" → $(\mu_0/4\pi)qv\sin\theta/r^2$
"long wire B" → $\mu_0I/(2\pi r)$
"inside wire B (r<R)" → $\mu_0Ir/(2\pi R^2)$
"center loop B" → $\mu_0I/(2R)$
"on axis loop B" → $\mu_0IR^2/[2(R^2+z^2)^{3/2}]$
"torque on loop" → $\mu\times B=NIAB\sin\theta$
"PE of dipole in B" → $-\mu\cdot B$
"dI/dt in solenoid → E" → $(\mu_0nr/2)(dI/dt)$
"charging cap → B" → $\mu_0I_C\,r/(2\pi R^2)$
"displacement current" → $I_D=\varepsilon_0\,d\Phi_E/dt$
"rotating coil ε" → $NBA\omega\sin(\omega t)$
"max EMF rotating" → NBAω
"flux when B(t)" → $\varepsilon=-A\cos\theta\,dB/dt$
"flux when θ(t)" → $\varepsilon=BA\sin\theta\,d\theta/dt$
"L of solenoid" → μ_0N^2A/ℓ
"L of toroid" → $\mu_0N^2A/(2\pi r)$
"inductor energy" → $\frac{1}{2}LI^2$
"u_B (B-field density)" → $B^2/(2\mu_0)$
"RL energizing" → $(\varepsilon/R)[1-e^{-Rt/L}]$
"RL decay" → $I_0e^{-Rt/L}$
"RL find L" → $L=Rt/\ln(I_0/I)$
"LC freq" → $1/(2\pi\sqrt{LC})$
"LC ω₀" → $1/\sqrt{LC}$

"LRC critical R" → $2\sqrt{L/C}$
"LRC underdamped ω'" → $\sqrt{1/LC-R^2/4L^2}$
"amplitude decay LRC" → $e^{-Rt/2L}$
"X_L" → ωL
"X_C" → 1/(ωC)
"impedance Z" → $\sqrt{R^2+(X_L-X_C)^2}$
"resonance ω" → $1/\sqrt{LC}$
"phase angle" → $\tan\varphi=(X_L-X_C)/R$
"P_avg AC" → I_{rms}^2R
"V_rms" → $V_{max}/\sqrt{2}$
"transformer V" → $V_2/V_1=N_2/N_1$
"transformer I" → $I_2/I_1=N_1/N_2$
"EM wave B from E" → B=E/c
"intensity from E_m" → $\frac{1}{2}c\varepsilon_0E_m^2$
"Poynting" → $(1/\mu_0)E\times B$
"|S|" → $EB/\mu_0=E^2/(c\mu_0)$
"rad pressure absorb" → I/c
"rad pressure reflect" → 2I/c
"k from λ" → 2π/λ
"ω from f" → 2πf
"c from μ₀,ε₀" → $1/\sqrt{\mu_0\varepsilon_0}$
"E and B in phase" → both sin(kx−ωt)
"Maxwell ∮E·dA" → Q_{encl}/ε_0
"Maxwell ∮B·dA" → 0
"Maxwell ∮E·dl" → $-d\Phi_B/dt$
"Maxwell ∮B·dl" → $\mu_0(I_C+\varepsilon_0\,d\Phi_E/dt)$
"speed in medium" → c/n
"f×λ" → c (or v in medium)
"E density u_E" → $\frac{1}{2}\varepsilon_0E^2$